

$$P(A) \times P(B) = \{\emptyset, (\{1\}, \{3\}), (\{1\}, \{4\}), (\{1\}, \{3, 4\}), (\{2\}, \{3\}), (\{2\}, \{4\}), (\{2\}, \{3, 4\}), (\{1, 2\}, \{3\}), (\{1, 2\}, \{4\}), (\{1, 2\}, \{3, 4\})\}$$

$\therefore P(A \times B) \neq P(A) \times P(B)$ as we have given a counterexample. $\square$

44) Prove or disprove that if A , B and C are nonempty sets and $A \times B = A \times C$, then $B = C$.

Solution: By Defn. 2, Both $A \times B$ and $A \times C$ contain the same elements.

Let, $(x, y) \in A \times B$. By Defn. 8, $x \in A$ and $y \in B$.

$\therefore (x, y) \in A \times C$. By Defn. 8, $y \in C$.

~~Since (x, y) is an arbitrary element of $A \times B$.~~
 $A \times B$ contains all such ordered pairs. $\therefore$ By Defn. 3, $B \subseteq C$.

Let (p, q) be an arbitrary element of $A \times C$. By Defn. 8, $p \in A$ and $q \in C$.

By Defn. 2, $(p, q) \in A \times B$. By Defn. 8, $q \in B$.

$\therefore$ By Defn. 3, $C \subseteq B$. $\therefore B = C$ $\square$

49) The defining property of an ordered pair is that two ordered pairs are equal if and only if their first elements are equal and their second elements are equal. Surprisingly, instead of taking the ordered pair as a primitive concept, we can construct ordered pairs using basic notions from set theory. Show that if we define the ordered pair (a, b) to be $\{\{a\}, \{a, b\}\}$, then $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

Solution: Let, $(a, b) = (c, d)$. $\therefore \{\{a\}, \{a, b\}\}$ and $\{\{c\}, \{c, d\}\}$ are two equal sets. By Defn. 2, both these sets must contain the same elements.

$$|\{a\}| = 1 \quad |\{c\}| = 1 \quad |\{a, b\}| = 2 \quad |\{c, d\}| = 2$$

$\therefore \{a\}$ and $\{c\}$ must be equal sets and $\{a, b\}$ and $\{c, d\}$ must be equal sets. $\therefore$ By Defn. 2, $a = c$. Also by defn. 2, The elements of $\{a, b\}$ and $\{c, d\}$ must be the same. $\therefore a = c$, $\therefore b = d$